THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · EASY LEVEL CHAPTER TEST SET 3

Chapter 4: **Carbon and its Compounds** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Vinegar contains: 1

(a) methanol **(b)** ethanol **(c)** ethene **(d)** ethanoic acid

2. The formula of ethanol is: 1

(a) C_2H_6 **(b)** CH_3OH **(c)** C_2H_5OH **(d)** CH_3COOH

3. The functional group of aldehydes is: 1

(a) -COOH **(b)** -CHO **(c)** -OH **(d)** -Cl

4. Esters have a: 1

(a) no smell **(b)** pungent smell **(c)** rotten smell **(d)** fruity smell

5. The formula of ethanoic acid is: 1

(a) CH_3COOH **(b)** CH_3CHO **(c)** C_2H_5OH **(d)** $HCOOH$

6. Soap is a: 1

(a) sodium salt of fatty acid **(b)** alcohol

(c) ester **(d)** hydrocarbon

7. Catalyst used in hydrogenation of oil is: 1

(a) nickel **(b)** copper **(c)** zinc **(d)** iron

8. A covalent bond is formed by: 1

(a) loss of electrons **(b)** sharing of electrons

(c) gain of electrons **(d)** transfer of electrons

9. The molecular formula of butane is: 1

(a) C_4H_{10} **(b)** C_4H_8 **(c)** C_3H_8 **(d)** C_4H_6

10. Detergents are better than soaps in: 1

(a) distilled water **(b)** rainwater **(c)** hard water **(d)** soft water

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What is the use of soap? 2

12. What are unsaturated hydrocarbons? Give an example. 2

13. What is catenation? 2

14. Name the gas evolved when ethanoic acid reacts with sodium carbonate. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Write the IUPAC names of CH_3OH, C_2H_5OH and C_3H_7OH. 3

16. Write the equation of the reaction of ethanol with sodium. 3

17. Why is diamond used in cutting glass? 3

OR

Write the IUPAC names of $HCOOH$, CH_3COOH and C_2H_5COOH. 3

18. Write the equation of hydrogenation of ethene and name the catalyst. 3

OR

Why is graphite used for electrodes and as a lubricant? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Write the names and structures of two isomers of butane. State the number of isomers of pentane. 5

OR

Write five differences between ethanol and ethanoic acid (formula, group, litmus, reaction with sodium carbonate, use). 5

20. Describe the reaction of ethanol with sodium, with alkaline $KMnO_4$ and with conc. H_2SO_4. Give products. 5

OR

Name the functional group and one example for halo, alcohol, aldehyde, ketone and carboxylic acid. 5

END OF QUESTION PAPER — Total: 40 Marks

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